SYSTEM ARCHITECTURE & CODE MANUAL

AmazingDeal (amazingdeal.co.in)

The Cloud-Native Multi-Tenant Hyperlocal E-Commerce Platform for Indian Retailers. Complete Engineering Walkthrough from Design to Code.

Spring Boot 3.3.5 • Java 21 LTS • Neon PostgreSQL • React 18 • Cloudflare Pages/Workers/R2

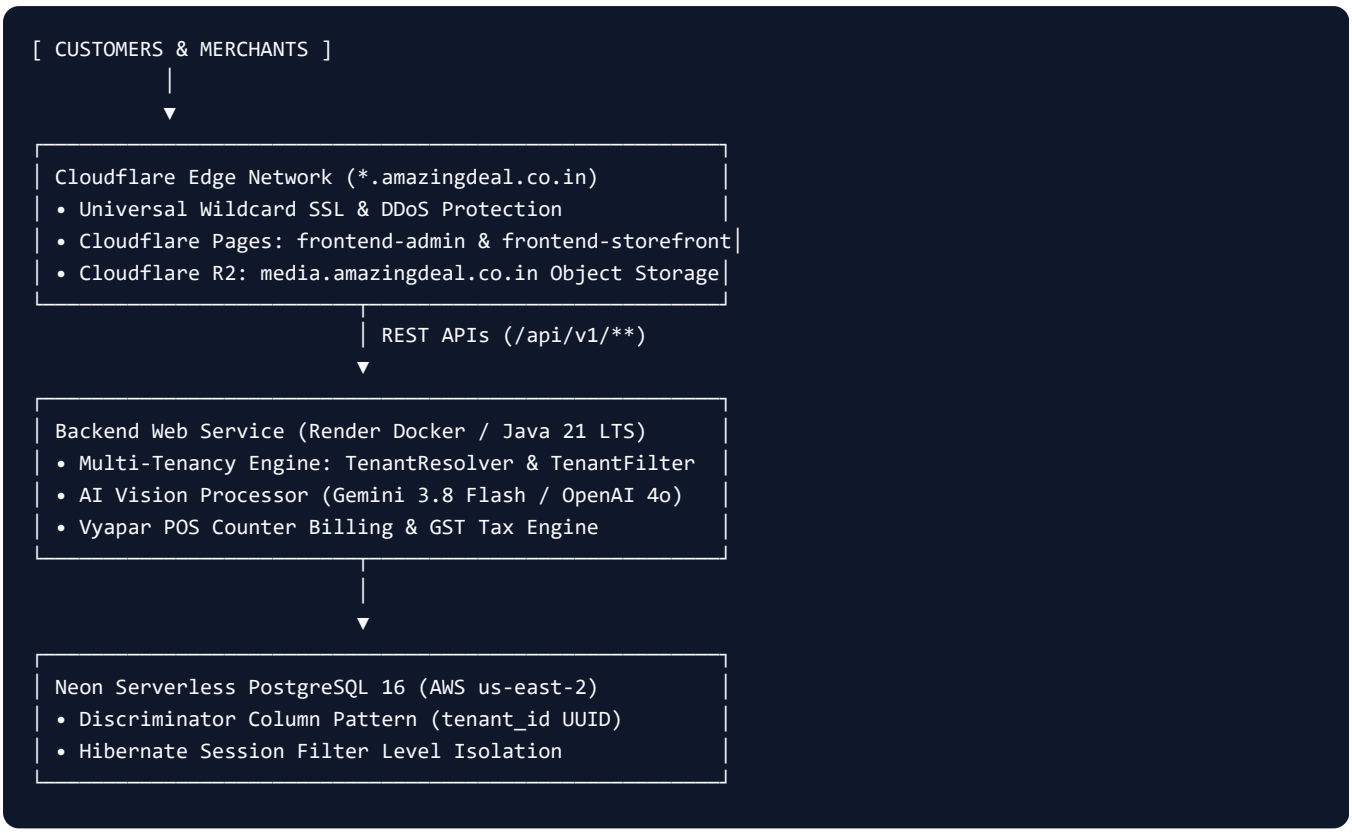
Table of Contents

1. **Executive Summary & High-Level System Architecture** (Topology, Tech Stack, Cloud Free-Tier)
2. **Multi-Tenancy Engine & Request Lifecycle** (Discriminator Column, ThreadLocal Contexts, Hibernate Filters)
3. **Database Architecture & Entity-Relationship Schema** (9 Master Tables, Field Mappings, Indexing)
4. **Authentication, Security & RBAC Engine** (Multi-Role JWTs, Firebase Auth, Mandatory Checkout Guard)
5. **AI Product Studio & Vision Pipeline** (Gemini 3.8 Flash, OpenAI 4o-mini, 360° Turnaround Turntable)
6. **Indian Payment Stack & Hardware POS Printing** (0% Fee Direct UPI QR, Razorpay BYOK, ESC/POS, GST Invoicing)
7. **Logistics Engine & Delivery Fleet** (Shiprocket Domestic BYOK, Store Fleet Dispatch, Time Slots)
8. **Superadmin Master Operations Suite** (JPA Specifications, Merchant Controls, Impersonation Engine)
9. **Complete File-by-File Catalog & Code Walkthrough** (All 99 Backend + 51 Frontend Files Detailed)
10. **Cloud DevOps & Zero-Cost Infrastructure Playbook** (Render, Neon, Cloudflare, OCI ARM VM)

1. Executive Summary & High-Level System Architecture

AmazingDeal is engineered to give every Indian kirana store, boutique, restaurant, pharmacy, and diagnostic lab their own independent e-commerce presence without monthly subscription traps or 15-30% commissions.

High-Level System Topology



Core Technology Inventory

Tier	Technology	Version	Key Architectural Responsibility
Backend Core	Spring Boot	3.3.5	REST API engine, dependency injection, and JPA repository management
Runtime	Java (Eclipse Temurin)	21 LTS	Virtual threads, G1 Garbage Collector, container CGroup memory ergonomics
Database	Neon PostgreSQL	16	Serverless PostgreSQL with SSL connection pooling and branchable storage
Multi-Tenancy	Hibernate Filters	6.x	Automatic row-level discriminator isolation (`tenant_id = :tenantId`)
AI Vision	Google Gemini & OpenAI	3.8 Flash / 4o	Camera-to-listing attribute extraction and 360° photoshoot generation
Admin Frontend	React + TypeScript	18.3.1	Merchant dashboard, POS counter billing, GST reporting, and Superadmin suite
Storefront	React + Firebase	18.3.1 / 13.0	Customer storefront, Google 1-Tap authentication, cart, and 1-step checkout
Media Delivery	Cloudflare R2	S3 API	Product catalog photos and QR sticker storage with 0 egress bandwidth fees

2. Multi-Tenancy Engine & Request Lifecycle

AmazingDeal uses the **Discriminator Column Pattern** with automated Hibernate session filtering. Every multi-tenant entity stores a `tenant_id UUID` column.

How Request Isolation Works (Step-by-Step):

1. **Subdomain Extraction (`TenantResolver.java`):** Inspects the HTTP `Host` header. E.g., for `sharma.amazingdeal.co.in`, it extracts `sharma`. Fallbacks check `?store=sharma` or header `X-Tenant-ID`.
2. **Tenant Validation & Context Binding (`TenantFilter.java`):** Resolves the tenant from database. If active, binds the UUID to ThreadLocal `TenantContext.setTenantId(tenant.getId())`.
3. **Hibernate Session Injection:** The filter enables Hibernate's session filter:

   ```
   session.enableFilter("tenantFilter").setParameter("tenantId", tenantId);
   ```

   Every JPA query subsequently executed by repositories automatically appends `WHERE tenant_id = 'xxxx'`. It is mathematically impossible to leak data across stores.
4. **Audit Assignment (`TenantAwareEntity.java`):** When creating records, `@PrePersist` automatically reads `TenantContext.getTenantId()` and injects it before the SQL INSERT statement runs.
5. **Thread Cleanup:** A `finally` block executes `TenantContext.clear()` to prevent ThreadLocal memory leakage across thread pool executions.

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3. Database Architecture & Entity Schema

The database schema consists of 9 core tables structured for hyperlocal multi-tenancy and high performance:

Table Name	Key Columns	Isolation	Purpose
`tenants`	id, store_name, subdomain, gstin, state_code, upi_vpa, subscription_plan	Global Table	Master store registrations, credentials, and business settings
`products`	id, tenant_id, title, price, compare_price, category, inventory, hsn_sac, gst_rate, images_json	Tenant-Scoped	Merchant product catalog, inventory tracking, and tax classifications
`orders`	id, tenant_id, customer_id, order_number, total_amount, payment_method, order_status, items_json	Tenant-Scoped	Customer orders and in-store POS counter bills
`customers`	id, firebase_uid, full_name, phone, email, status	Global Registry	Registered Indian buyers across all stores
`customer_addresses`	id, customer_id, address_type, street_address, pincode, landmark, is_default	Customer-Scoped	Saved multi-address book for frictionless checkout
`parties`	id, tenant_id, party_name, party_type, gstin, phone, current_balance, credit_limit	Tenant-Scoped	Vyapar-grade customer and supplier Udhaar Khata ledgers
`invoices`	id, tenant_id, party_id, invoice_number, document_type, taxable_amount, cgst, sgst, igst, total	Tenant-Scoped	Official GST tax invoices and estimates with state tax splitting
`superadmins`	id, email, password_hash, full_name, role, status, last_login_at	System Admin	Platform master administrators with global oversight
`razorpay_transactions`	id, order_id, payment_id, amount, plan_id, subdomain, status	Audit Ledger	Audit trail of platform SaaS subscription payments

4. Authentication, Security & RBAC Engine

AmazingDeal enforces defense-in-depth across three separate user roles using stateless JJWT:

- `ROLE_MERCHANT` : Store owners administering products, orders, POS billing, and GST reports.
- `ROLE_CUSTOMER` : Shoppers logging in via Firebase Authentication (Google 1-Tap or Mobile SMS OTP). Mandatory at checkout to prevent guest order fraud.
- `ROLE_SUPERADMIN` : Master platform operators overseeing stores, platform revenue, and server health.

> **Mandatory Customer Checkout Guard:** Both frontend (`CartDrawer.tsx` , `CheckoutModal.tsx`) and backend (`StorefrontController.placeOrder`) strictly validate customer identity. Unauthenticated requests are rejected with `HTTP 401 Unauthorized` .

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5. AI Product Studio & Vision Pipeline

Merchants can snap a single packaging photo. `AiCatalogGeneratorService.java` calls Google Gemini 3.8 Flash (with automatic OpenAI GPT-4o-mini fallback) to extract:

- Clean Title, Brand Name, and Subcategory
- Recommended Selling Price and Compare MRP
- Detected Barcode / EAN number
- HSN / SAC Code and matching Indian GST tax slab (0%, 5%, 12%, 18%, 28%)
- **360° Studio Enhancer:** Generates 4 synthetic angles (0° Front Hero, 45° Perspective, 90° Side Profile, Macro Detail) displayed in an interactive touch/mouse turntable scrubber.

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6. Indian Payment Stack & Hardware Printing

1. **Direct UPI Dynamic QR (0% MDR):** Formats NPCI standard deep link: `upi://pay?pa=store@okaxis&pn=StoreName&am=150.00&tn=ORD-1001&cu=INR` Generates instant QR code via Google ZXing library. Customers tap to open GPay, PhonePe, or Paytm.
2. **Razorpay BYOK:** Merchants can plug in their own Razorpay Key ID and Secret. Backends enforce strict cryptographic HMAC-SHA256 signature verification with zero mock

bypasses.

3. **Vyapar POS Counter Billing (`PosCounter.tsx`):** High-speed billing with barcode scanner support, hotkeys (`F2` New Bill, `F4` Quick Cash, `F8` Clear), and automated Cash Change Calculator (₹100, ₹200, ₹500 rupee presets).
4. **Thermal Printing:** Native 58mm and 80mm ESC/POS monospace receipt formatting.
5. **Isolated Print Engine (`printInvoices.ts`):** Renders A4 GST Tax Invoices in a dedicated hidden iframe, completely eliminating browser modal clipping and ensuring zero blank sheets on printers like Epson L3310.
6. **GSTR-1 Excel Engine (`gstExport.ts`):** Generates 4-sheet Microsoft Excel XML Workbooks (.xls) for Master Sales Register, Table 4 (B2B), Table 7 (B2C), and Table 12 (HSN Summary).

7. Logistics Engine & Delivery Fleet

- **Shiprocket Domestic Courier (BYOK):** Automated AWB generation, shipping labels, and live serviceability check across 29,000+ Indian pincodes.
- **Store Rider Dispatch:** Hyperlocal radius calculation, whitelisted pincodes, delivery time slot selection ("Today Evening", "Rush Express 60-mins"), and 1-tap WhatsApp dispatch link with customer address, Google Maps GPS coordinates, and cash-to-collect instructions.

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8. Superadmin Master Operations Suite

Accessed at `/superadmin` and housed inside `frontend-admin`:

- **JPA Dynamic Specifications:** Uses `JpaSpecificationExecutor<T>` and Criteria API predicates, completely eliminating untyped null parameter JDBC crashes in PostgreSQL.
- **Platform KPIs:** Live metrics for Active/Suspended Merchants, Customers, Orders, Platform GMV, and SaaS Revenue.
- **1-Click Store Impersonation:** Generates scoped merchant JWTs allowing superadmins to inspect any store's dashboard with a sticky top return banner.
- **JVM Memory Meter:** Real-time telemetry of committed heap, active memory, and container headroom.

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9. Complete File-by-File Code Walkthrough

Backend Core & Multi-Tenancy

File Path	Key Class	Detailed Purpose
`multitenancy/TenantResolver.java`	TenantResolver	Parses HTTP Host header for subdomain, query param `?store=`, or `X-Tenant-ID`.
`multitenancy/TenantFilter.java`	TenantFilter	Spring filter validating tenant and binding Hibernate session filter parameter.
`multitenancy/TenantContext.java`	TenantContext	ThreadLocal holder for current request's tenant UUID and subdomain.
`multitenancy/CustomerContext.java`	CustomerContext	ThreadLocal holder for authenticated customer UUID.
`multitenancy/TenantAwareEntity.java`	TenantAwareEntity	Base entity class with `@PrePersist` automatic tenant assignment.
`config/SecurityConfig.java`	SecurityConfig	Configures stateless Spring Security 6 filter chain and endpoint RBAC.
`config/JwtFilter.java`	JwtFilter	Validates incoming Bearer JWT tokens and sets Spring SecurityContext.
`config/JwtTokenProvider.java`	JwtTokenProvider	Generates HMAC-SHA256 tokens for Merchants, Customers, and Superadmins.

Backend Feature Modules

Module	Key Files	Detailed Purpose
AI Vision	`AiCatalogController.java`, `AiCatalogGeneratorService.java`	Camera-to-listing extraction and 360° photo synthesizer using Gemini and OpenAI.
Billing & Tax	`BillingController.java`, `BillingService.java`, `Invoice.java`	GST tax invoicing, state tax split logic, HSN summary, and GSTR-1 CSV export.
Customers	`CustomerAuthController.java`, `CustomerService.java`	Firebase authentication token exchange and customer address book management.
Orders & POS	`StorefrontController.java`, `AdminOrderController.java`, `OrderService.java`	Order placement with login guard, inventory decrement, and ESC/POS thermal formatting.
Parties & Khata	`PartyController.java`, `PartyService.java`, `Party.java`	Customer/Supplier Udhaar Khata ledgers with live receivables/payables tracking.
Payments	`RazorpayController.java`, `RazorpayService.java`	Razorpay order creation, payment capture, and HMAC-SHA256 webhook verification.
Products	`AdminProductController.java`, `ProductService.java`	Catalog CRUD, stock status toggling, and multi-image JSONB gallery management.
Logistics	`ShiprocketService.java`	Domestic courier BYOK token caching, AWB dispatch, and pincode serviceability.
Superadmin	`SuperadminController.java`, `SuperadminService.java`	Global cross-tenant analytics, store management, and impersonation engine.

Frontend Admin Suite (`frontend-admin/src`)

Component / Utility	Key Responsibility
`App.tsx`	Master dashboard state coordinator, role-aware routing, and periodic data refresh.
`PosCounter.tsx`	Instant counter POS billing with barcode scanner support, hotkeys, and change calculator.
`InvoiceList.tsx`	Tax invoice register with period filters (This Month, Last Month) and GST KPI cards.
`GstInvoiceModal.tsx`	A4/A5 professional GST tax invoice with print and 1-tap WhatsApp sharing.
`BatchInvoicesModal.tsx`	Batch print & PDF export with manual date pickers and multi-attribute search.
`PartiesKhata.tsx`	Customer and supplier credit ledgers with 1-tap WhatsApp payment reminders.
`StoreSettings.tsx`	5-tab studio for Bank IFSC auto-fetch, 15-digit GSTIN parser, and BYOK credentials.
`AiProductScannerModal.tsx`	AI camera studio with drag rotation turntable and studio command bar.
`superadmin/*`	Full Superadmin management suite (Overview, Stores, Customers, Orders, Subscriptions, Health).
`utils/printInvoices.ts`	Zero-blank-page isolated iframe print engine for thermal, inkjet, and laser printers.
`utils/gstExport.ts`	Multi-sheet Microsoft Excel XML (.xls) GSTR-1 workbook generator.

Frontend Storefront Suite (`frontend-storefront/src`)

Component / Hook	Key Responsibility
`MerchantLandingPage.tsx`	Apex domain (`amazingdeal.co.in`) marketing landing page with instant store wizard.
`StoreHeader.tsx`	Branded store header with category navigation and customer login trigger.
`ProductCard.tsx`	Responsive product card with discount badges, stock indicators, and quick-add button.
`CartDrawer.tsx`	Slide-out cart drawer with delivery fee calculation and mandatory customer auth guard.
`CheckoutModal.tsx`	1-step checkout with saved address picker, delivery time slots, and payment modes.
`CustomerAuthModal.tsx`	Firebase Google 1-Tap and Mobile SMS OTP customer sign-in modal.
`hooks/useSubdomain.ts`	Extracts current store subdomain from URL hostname or query parameter.
`hooks/useCustomerAuth.ts`	Customer Firebase auth state listener and customer JWT token persistence.

10. Cloud DevOps & Zero-Cost Infrastructure Playbook

1. Production Deployment Topology

- **Database:** Neon Serverless PostgreSQL 16 (AWS us-east-2). Automatically scales and sleeps when inactive to maintain ₹0 hosting cost.
- **Backend Web Service:** Render Free Tier (Docker runtime, Java 21 LTS). Paired with an automated Keep-Alive ping (Cron-job.org / UptimeRobot) pinging `/api/v1/health` every 10 minutes to eliminate cold starts.
- **Frontend Static Assets:** Cloudflare Pages / Workers. Unlimited bandwidth, deployed via `wrangler deploy` with custom routes for `admin.amazingdeal.co.in` and `*.amazingdeal.co.in`.
- **Object Storage:** Cloudflare R2 S3-compatible bucket (`media.amazingdeal.co.in`) with 10 GB free storage and 0 egress fees.

2. Alternative OCI Always-Free Stack

The repository includes turnkey configuration for an **Oracle Cloud Always-Free ARM VM** (4 OCPUs, 24 GB RAM, 200 GB SSD) via `docker-compose.oci.yml`:

```
# Start complete local stack on OCI VM
docker compose -f docker-compose.oci.yml up -d --build
```

Includes local PostgreSQL 16 container, Spring Boot backend container, and Caddy container providing automated Let's Encrypt SSL certificates for `api.amazingdeal.co.in`.
